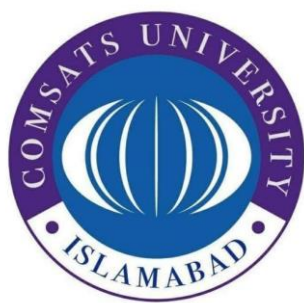




**COMSATS University Islamabad**  
**Lahore Campus**

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| Reg..No:        | SP25-BSE-013,SP25-BSE-001,SP25-BSE-067            |
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| Course Title    | DSA                                               |
| Section         | A                                                 |
| Assignment#     | Project Proposal                                  |
| Submitted To    | Mrs.Humera Faisal                                 |
| Submission Date | 30 April, 2026                                    |

## PROJECT PROPOSAL

# Hostel Kharcha Manager

## 1. Problem Statement

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Hostel life is a defining chapter of a student's academic journey but it also presents a routine but unavoidable problem: managing expenses. Each month, housemates combine their resources to purchase groceries, pay for electricity, internet, and other essentials. What starts out as a straightforward pooling of resources quickly becomes an unregulated debt matrix involving complex accounting, and the attendant "rona dhona" over the loose change. One person can't recall who took out groceries last week; another claims they have already paid their share of the electricity bill. Paper-based or even digital ledger keeping, in notebooks or WhatsApp messages, is vulnerable, irregular and non-auditable. Manual calculations are prone to human error, which leads to a breakdown in trust and genuine disputes. There is an urgent need for a systematic, accurate, efficient solution to record all transactions, track outstanding amounts, and provide summaries at will, removing all doubts, restoring justice and peace in the financial battlefield of hostel life.

## 2. Technical Implementation

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This project is implemented as a console-based application in C++, with a deliberate focus on low-level programming principles central to the study of Data Structures and Algorithms. The entire system is built upon manual pointer manipulation and dynamic memory management using 'new' and 'delete' operators, ensuring that each node, edge, and element is explicitly allocated and deallocated by the programmer.

All data structures including the Hash Map, Doubly Linked List, Stack, and Heap are designed and implemented from scratch. This approach reinforces a deep understanding of memory layout, algorithmic complexity, and the engineering trade-offs inherent in real-world software design.

## 3. Features of the Program

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The following six core features constitute the functional scope of the Hostel Kharcha Manager:

### Feature 1: Member Profile Management

Each hostel member is registered in the system with a unique ID, name, and running balance. Profiles can be created, updated, and retrieved instantly at any point during program execution. The system ensures no duplicate entries and maintains data integrity across all operations.

**Data Structure:** Custom Hash Map achieving  $O(1)$  average-time insertion and lookup per member profile.

### Feature 2: Transaction Logging

Every expense whether a shared meal, a utility bill, or a one-off purchase is recorded as a transaction node with a timestamp, amount, payer, and list of involved members. The log preserves the complete chronological history of all financial activity within the hostel.

**Data Structure:** *Doubly Linked List enabling efficient forward and backward traversal of the transaction timeline.*

### Feature 3: Undo Functionality

Mistakes in data entry are inevitable. The undo feature allows users to instantly revert the most recently added transaction, restoring all affected balances to their prior state. Multiple consecutive undos can be performed to roll back a series of erroneous entries.

**Data Structure:** *Stack (LIFO Last In, First Out) the most recent transaction is always at the top for  $O(1)$  reversal.*

### Feature 4: Automated Bill Splitting

When a shared expense is logged, the system automatically calculates each member's proportional share and updates their individual balances accordingly. Both equal splits and custom weighted distributions are supported, eliminating the need for manual arithmetic entirely.

**Data Structure:** *Iterative traversal of the Member Hash Map ensuring all relevant profiles are updated in a single pass.*

### Feature 5: Debt Prioritization

The system maintains a live ranking of members by the amount they currently owe. This feature allows the group to immediately identify the highest debtor and prioritize settlement, preventing old debts from being overlooked. The ranking updates dynamically after every transaction.

**Data Structure:** *Max-Heap (Priority Queue) the member with the highest outstanding balance is always extractable in  $O(\log n)$ .*

### Feature 6: Summary Reporting

At any time, the user can generate a complete financial summary displaying each member's total expenses paid, total share owed, and net balance (positive for credit, negative for debt). The report provides a clear, audit-ready ledger of the hostel's financial state.

**Data Structure:** *Linear traversal of the Doubly Linked List and in-order traversal of balance records for a sorted ledger view.*

## 4. Expected Outcomes

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- A fully functional, menu-driven C++ console application.
- Practical demonstration of at least four distinct data structures: Hash Map, Doubly Linked List, Stack, and Max-Heap.
- Transparent memory management with no reliance on STL containers.
- A real-world problem solved through the principled application of algorithmic thinking.
- Complete source code documentation and a final project report.